A Fuzzy Network for Medical Insurance Risk Classification

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Abstract—**This paper presents a cascaded fuzzy network for the binary classification of medical insurance charge risk, implemented using MATLAB Fuzzy Logic Toolbox and Fuzzy Network Toolbox. 2 Mamdani Type-1 fuzzy inference systems are constructed, then a cascaded rule base is formed via the horizontal_merging function, combining six smoker–age input combinations with a charge category mapping through Boolean matrix multiplication. The system is evaluated on a publicly available dataset of 1,337 medical insurance records. On a held-out test set of 267 records, the combined fuzzy network achieves 91.8% accuracy, with a High-class F1 score of 0.820, a precision of 90.9%, and a recall of 74.6%. Performance is comparable to the Gradient Boosting benchmark of 92.0% and surpasses the Random Forest benchmark of 83.4%. Analysis shows that smoking status is the primary predictor of insurance charges, supporting the modular decomposition strategy.**

Keywords—fuzzy network, medical insurance, machine learning

I. Introduction

The accurate assessment of medical insurance risk is an important challenge for healthcare insurers, policymakers, and actuarial practitioners. Such an evaluation system can also be used to analyse citizens' health levels and life expectancy. Accurate risk stratification enables insurers to price policies fairly, distribute resources efficiently, and identify high-risk individuals who may benefit from preventive interventions. Traditional actuarial models rely on linear assumptions and crisp categorical boundaries that are unable to capture the inherently imprecise and uncertain nature of human health data. A patient does not transition abruptly from being 'young' to 'middle-aged', nor does a body mass index of 29.9 represent a categorically different health profile from one of 30.1. This linguistically based uncertainty is a fundamental characteristic of medical risk assessment that classical statistical models are not designed to handle.

Fuzzy logic, introduced by Zadeh [1], provides a mathematical model for representing and reasoning with language uncertainty via graded membership functions. Unlike machine learning models such as random forests or gradient boosting, fuzzy systems produce decisions that can be expressed in interpretable IF-THEN rules directly readable by domain experts. This property has motivated substantial research into fuzzy approaches for insurance risk modelling, as demonstrated by Shapiro [3] and Milosevic et al. [4].

However, previous fuzzy networks for insurance usually employed a single, flat rule-based model. As the number of input variables increases, the number of rules grows exponentially with the number of input variables [5]. This made the rule base difficult to design, validate and maintain. Such scaling issues have limited the practical application of fuzzy logic in multidimensional medical risk assessment. This paper addresses this scalability limitation by proposing a cascaded fuzzy network architecture for the classification of medical insurance costs, implemented using both the MATLAB Fuzzy Logic Toolbox and the Fuzzy Network Toolbox. The fuzzy network breaks down the risk assessment problem into two semantically independent subsystems: a health risk subsystem that captures physiological risk factors (age and body mass index (BMI)), and a lifestyle risk subsystem that captures behavioural risk factors (smoking status and number of dependants). Subsequently, these two subsystems are connected via a horizontal_merging function to form a cascaded network, which propagates reasoning from one rule base to the next. The proposed fuzzy network is evaluated on a publicly available medical insurance dataset [7] comprising 1,337 records with 7 features, including a continuous cost output.

This paper is organised as follows: Section II reviews the relevant literature on fuzzy logic theory, its application to insurance and medical risk assessment, and the machine learning benchmarks used for comparison. Section III describes the suggested methodology, including the dataset, sub-system design and network merging procedure. Section IV presents the experimental results. Section V evaluates and discusses the findings relative to benchmark methods. Section VI concludes this paper.

II. Review

A. Theoretical Foundation of Fuzzy Logic

The mathematical basis of this work rests on the theory of fuzzy sets introduced by Zadeh [1], which extended classical set theory by permitting elements to have graded degrees of membership in the interval [0,1], rather than classical Boolean logic with only 0 and 1. In fuzzy logic, 1 indicates full membership, 0 indicates non-membership, and intermediate values represent partial membership. This formalism enables the representation of linguistic variables, such as 'age is Middle' or 'BMI is Overweight', as continuous, overlapping regions rather than as discrete, crisp boundaries.

Mamdani and Assilian [2] developed the first practical fuzzy logic controller based on this foundation. Employing a set of IF-THEN rules to evaluate the membership degrees. This model remains the dominant inference architecture in applied fuzzy systems due to its simple rule structure and understandability, and is employed in both subsystems of this work. This study uses MATLAB Fuzzy Logic Toolbox and implements the Mamdani Type-1 inference mechanism.

B. Fuzzy Logic in Insurance and Medical Risk Assessment

The application of fuzzy logic to the insurance problem has a well-established theoretical basis. Shapiro provided a comprehensive survey of fuzzy logic applications in insurance [3], demonstrating its practicality in representing the fundamental ambiguity of concepts such as 'high risk', 'acceptable claims' or 'critical health conditions'. The review notes that fuzzy systems are suitable for the insurance sector. Although expert knowledge exists in this field, it is difficult to apply precise numerical thresholds, and the costs of misclassification are asymmetric. Milosevic et al. [4] applied fuzzy logic directly to the risk classification of insurance clients, combining the Fuzzy Analytic Hierarchy Process with Mamdani's model to assign risk grades based on multiple criteria, including financial history and demographic variables. Compared with traditional scoring methods, their system showed quantifiable improvements in classification consistency. However, it employed a single, flat rule base that aggregated all input variables without modular decomposition. This approach illustrates the previous state of the art in fuzzy insurance systems and provides a direct methodological benchmark for the positioning of the modular architecture in this study.

C. *Dimensionality and Modular Fuzzy Networks*

A limitation of conventional fuzzy inference systems is the exponential growth of the rule base with the number of input variables. For a system with n inputs, each partitioned into m linguistic terms, the number of rules for complete coverage is m^n. Ojha et al. [5] surveyed three decades of research on fuzzy system design and identified this rule explosion as a main obstacle to scalable fuzzy modelling, noting that hierarchical and modular decomposition strategies represent the most effective approach for managing this complexity and preserving system transparency. The theoretical solution to this problem was formalised by Gegov [6], who introduced the fuzzy network as a modular framework in which a complex inference problem is decomposed into a set of smaller, interconnected rule bases. Network operations including horizontal merging, vertical merging, and output merging enable rule bases to be combined systematically, with the horizontal_merging function performing Boolean matrix multiplication to eliminate intermediate variables from inference chain. This cascaded architecture eliminates intermediate variables from the final inference structure, diminishing computational cost while preserving the modular design rationale.

D. *Machine Learning Benchmarks for Insurance Cost Prediction*

A substantial body of machine learning research has applied supervised learning methods to the same medical insurance cost prediction problem addressed in this paper. Lantz [7], comprising 1,337 patient records, provided essential baseline comparisons for this study. Orji and Ukwandu [8] evaluated on a subset of 986 records drawn from the same Kaggle dataset with gradient boosting, random forests, and XGBoost. Their findings demonstrate that ensemble architectures successfully capture the nonlinear feature interactions (primarily smoking status) that drive the bimodal cost distribution. Nevertheless, continuous regression precision remains methodologically distinct from risk categorisation metrics.

For consistent evaluation on binary classification, Islam et al. [9] established baseline accuracies of 83.4% for random forests and 92.0% for gradient boosting using the same data. While high-performing, Rawat [10] highlights that such ensemble models present a major trade-off between predictive power and explainability. The opacity creates difficulties in regulated insurance markets where actuarial decisions must remain transparent to regulators and policyholders. Accordingly, this paper positions the proposed fuzzy network against these classification benchmarks using identical metrics, as summarised in Table I.

TABLE I. PERFORMANCE BENCHMARKS ON THE LANTZ MEDICAL INSURANCE DATASET.

Method	Task	Metric	Performance
Linear Regression (Orji & Ukwandu)	Regression	R^2	0.744
Random Forest (Orji & Ukwandu)	Regression	R^2	0.826
XGBoost (Orji & Ukwandu)	Regression	R^2	0.880
Random Forest (Islam et al)	Binary classif ication	Accuracy	83.4%
Gradient Boosting (Islam et al)	Binary classification	Accuracy	92.0%

III. METHODOLOGIES

A. *Dataset*

The dataset employed in this study is the medical insurance cost dataset compiled by Lantz [7]. The data comprises 1,337 records after removing 1 duplicate entry from the original 1,338 rows. Each record contains six input parameters, including age, sex, BMI, number of children, smoking status, and US region. A continuous output variable representing the annual insurance charge in US dollars. The dataset contains no missing values. Two features are excluded from the fuzzy system design to reduce the complexity for reasonable reasons. Sex exhibits negligible predictive power, with a mean charge difference of less than $1,400 between male and female policyholders. Region shows no statistically meaningful variation in mean charges across the four geographic zones. The remaining four features, age, BMI, smoking status, and number of children, are retained as inputs to the two sub-systems, with children clipped at three owing to the very low frequency of records with four or five dependants. The dataset is partitioned into a training set (80.0% of the original dataset) and a test set (20.0% of the original dataset). To ensure a stable split of the data, a fixed random seed was used. Data analysis of charge distributions throughout all input features reveals that the non-smoker subgroup has a mean annual charge of $8,434, whereas the smoker subgroup has a mean annual charge of $32,050. This property suggests a decision boundary at $16,000. Records with charges below $16,000 are designated the LowMedium class, and those at or above are designated the High class.

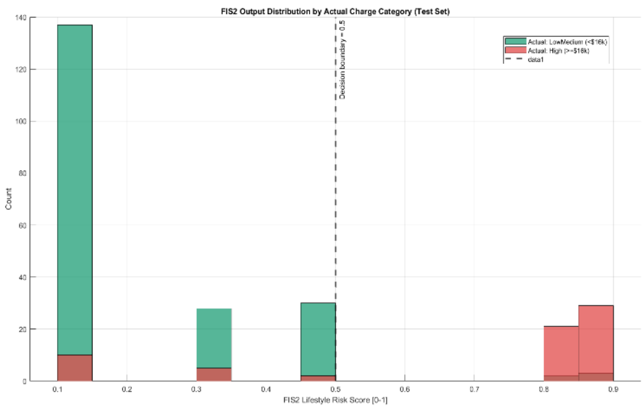

Fig. 1. Output Distribution by Actual Charge Category.

B. *Health Risk*

The first Mamdani Type-1 fuzzy inference system (FIS 1) models the physiological dimension of insurance risk. It takes age and BMI as inputs and produces a continuous body_risk score on the normalised interval [0,1]. The membership functions for age are defined as three triangular functions, Young, Middle, and Senior, with breakpoints grounded in the 33rd and 66th percentiles of the dataset age distribution (approximately 30 and 47 years, respectively). The BMI input employs three triangular membership functions aligned with WHO clinical thresholds: Normal [15,21,27] covering BMI below 25, Overweight [23,28,33] covering the 25 to 30 range, and Obese [29,55,55] covering BMI above 30. The body_risk output is described by three triangular membership functions, Low [0,0,0.35], Medium [0.2,0.5,0.75], and High [0.6,1,1]. 9 rules cover the full Cartesian product of the two inputs, as detailed in Table II. Rule consequents are assigned based on mean charge cross-tabulations: the Young-Normal and young-Overweight combinations yield mean charges below $7,500 and are therefore assigned to body_risk Low. All Middle-age combinations and the Senior -Normal combination with mean charges between $9,955 and $14,408 are mapped to body_risk of Medium. The rest are the Senior-Overweight and Senior-Obese combinations, with means of $15,340 and $18,729, respectively, and are assigned to body_risk High.

TABLE II. FIS 1 RULE BASE: AGE X BMI -> BODY_RISK.

	Rules
1	If age is Senior and BMI is Obese then body_risk is High
2	If age is Senior and BMI is Overweight then body_risk is High
3	If age is Young and BMI is Obese then body_risk is Medium
4	If age is Middle and BMI is Normal then body_risk is Medium
5	If age is Middle and BMI is Overweight then body_risk is Medium
6	If age is Middle and BMI is Obese then body_risk is Medium
7	If age is Senior and BMI is Normal then body_risk is Medium
8	If age is Young and BMI is Normal then body_risk is Low
9	If age is Young and BMI is Overweight then body_risk is Low

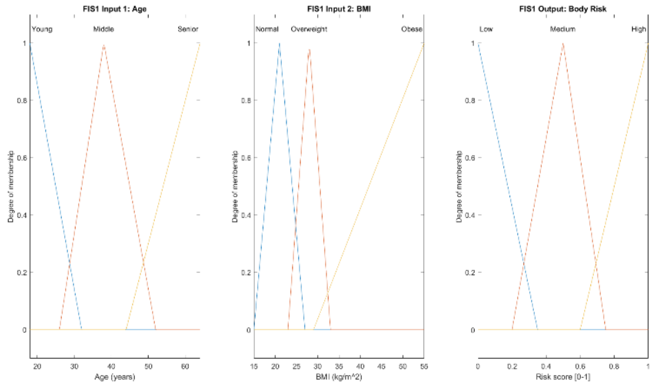

Fig. 2. FIS 1 Membership Functions: Age (left), BMI (centre), Body_Risk Output (right).

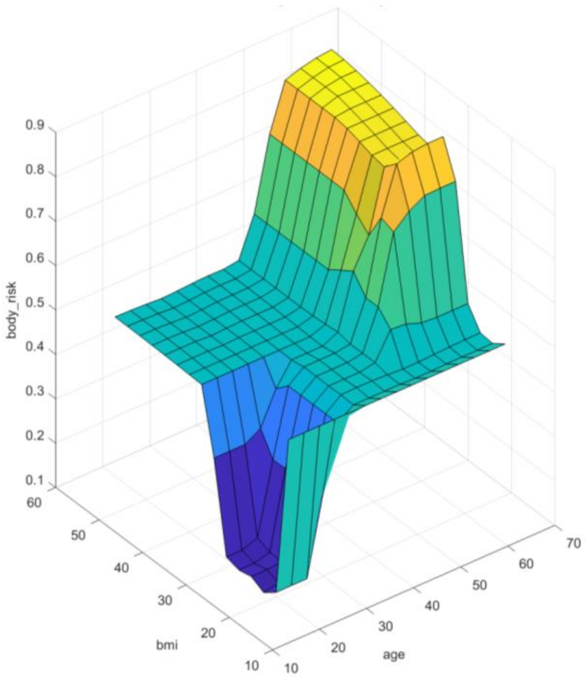

Fig. 3. Control Surface of FIS 1: Age and BMI verses Body_Risk Score.

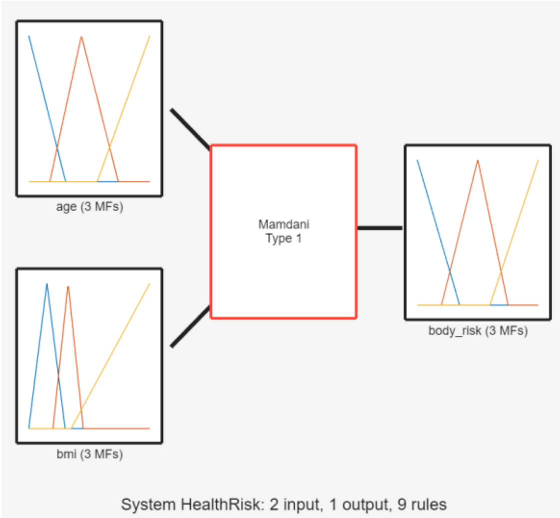

Fig. 4. Health Risk in Fuzzy Logic Designer.

C. *Lifestyle Risk*

The second Mamdani Type-1 fuzzy inference system (FIS 2) models the behavioural dimension of insurance risk. It takes smoking status and number of children as inputs and produces a continuous lifestyle_risk score on [0,1]. Smoking status is encoded as a binary numerical variable as 0 for non-smokers and 1 for smokers. It is described by two triangular membership functions, nonsmoker [0,0,0.5] and smoker [0.5,1,1]. The children's input is described by three triangular functions: None [0,0,1.5], Few [0.5,1.5,2.5], and Several [1.5,3,3], which represent the natural groupings in the dependent count distribution. 6 rules cover the full smoker x children input space. The data demonstrate that smoking status is the overwhelming determinant of charge level. All smoker subgroups range from $31,341 to $32,781, whilst all non-smoker subgroups range from $7,625 to $9,811. Hence, all smoker rules map to lifestyle_risk High, and the Nonsmoker rules map to Medium or Low, depending on the dependent count.

TABLE III. FIS 2 RULE BASE: SMOKER X CHILDREN -> LIFESTYLE_RISK.

	Rules
1	If smoker is Nonsmoker and children is None then lifestyle_risk is Low
2	If smoker is Nonsmoker and children is Few then lifestyle_risk is Low
3	If smoker is Nonsmoker and children is Several then lifestyle_risk is Medium
4	If smoker is Smoker and children is None then lifestyle_risk is High
5	If smoker is Smoker and children is Few then lifestyle_risk is High
6	If smoker is Smoker and children is Several then lifestyle_risk is High

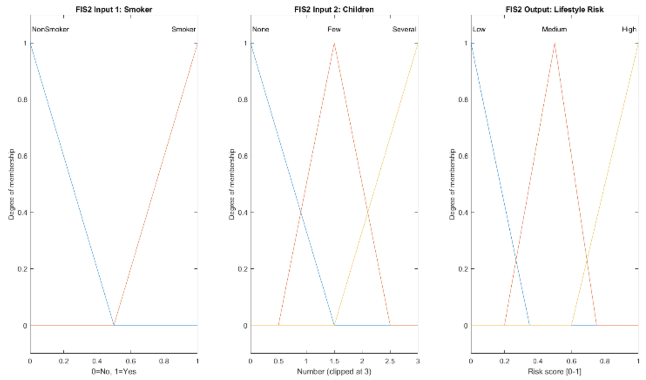

Fig. 5. FIS 2 Membership Functions: Smoking (left), Number of Children (centre), Lifestyle Risk Output (right).

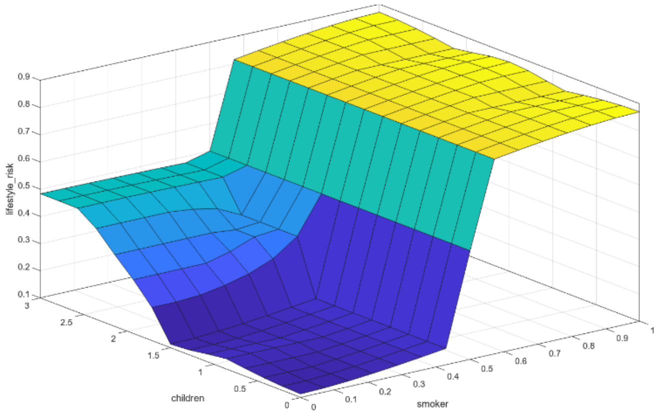

Fig. 6. Control Surface of FIS 2: Smoking and Number of Children verses Lifestyle Risk Score.

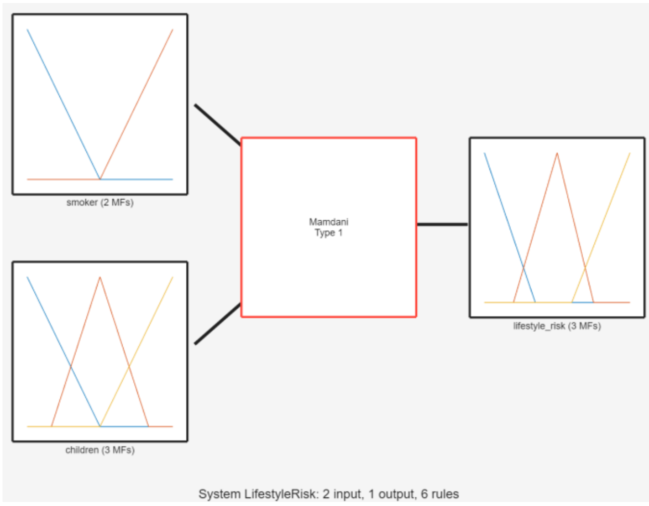

Fig. 7. Lifestyle Risk in Fuzzy Logic Designer.

D. *Cascaded Fuzzy Network*

The two subsystems are connected through the horizontal_merging function of the Fuzzy Network Toolbox [6]. Horizontal merging performs Boolean matrix multiplication of 2 rule bases, RB1 and RB2. The number of columns of RB1 must equal the number of rows of RB2. The input to the merged rule base is the same as the input of RB1, and the output from the merged rule base is the same as the output from RB2, with intermediate variables eliminated from the final inference structure. In this study, RB1 encodes the mapping from 6 smoker-age input combinations to 3 intermediate-risk outputs, producing a Boolean matrix of dimension 6 × 3. RB2 encodes the mapping from 3 risk index terms to 2 charge categories, producing a 3x2 matrix. The merged rule base is therefore a 6x2 matrix that maps directly from 6 smoker-age profiles to the charge category shown in Table IV. The design of RB1 is grounded in charge cross-tabulations: all non-smoker subgroups yield mean charges below $12,600 and are assigned a risk index of Low or Medium, whilst all smoker subgroups yield mean charges above $23,900 and are universally assigned a risk index of High. RB2 then maps Low and Medium risk index terms to the LowMedium charge category, and High risk index terms to the High charge category, consistent with the $16,000 decision boundary.

TABLE IV. MERGED RULE BASE (6X2).

Input Combination	LowMedium	High
Nonsmoker + Young	1	0
Nonsmoker + Middle	1	0
Nonsmoker + Senior	1	0
Smoker + Young	0	1
Smoker + Middle	0	1
Smoker + Senior	0	1

The full system architecture is illustrated in Fig. 8. FIS 1 and FIS 2 operate as independent Mamdani inference engines, each producing a continuous risk score that may be used in isolation or in combination. The cascaded network (RB1 -> RB2, merged by the horizontal_merging function) produces a discrete binary classification output from the smoker and age inputs alone. In the combined evaluation described in Section IV, the merged network prediction is used as the primary classifier; FIS 2 serves as a validation signal, and the two produce identical predictions on the test set, confirming the internal consistency of the network design.

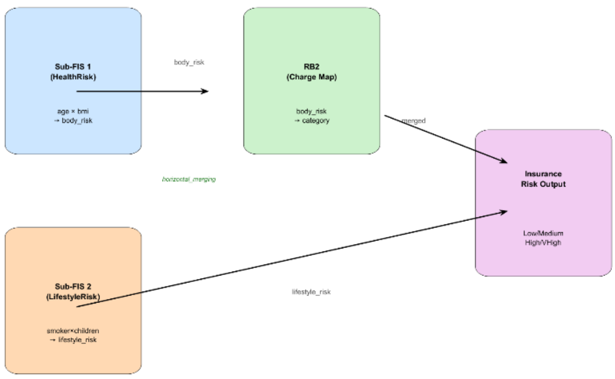

Fig. 8. Cascaded Fuzzy Network Architecture: FIS 1 (age, BMI -> body_Risk), FIS 2 (smoker, children -> lifestyle_risk), RB1 AND RB2 connected by horizontal_merging function to produce final charge category output.

E. *Evaluation Protocol*

System performance is assessed on the held-out test set of 267 records using overall classification accuracy, per-class precision and recall, and the F1 score for the minority High class. These metrics are selected to match those reported by the machine learning benchmarks in Table I, enabling direct quantitative comparison. The theoretical accuracy ceiling for any rule-based classifier on this dataset is analytically computed by identifying the majority charge class within each of the eighteen input-combination subgroups (smoker × age × BMI) and summing the resulting maximum correct classifications with an upper bound of 92.8%. This ceiling arises because a subset of 84 non-smoker records carries charges above $16,000 despite belonging to subgroups in which the overwhelming majority of charges fall below the threshold. Under the current subgroup partitioning strategy, no deterministic rule system can correctly classify these boundary cases without simultaneously misclassifying a larger number of correctly assigned records within the same subgroup.

IV. RESULTS

A. *Sub-system Performance*

Each component of the fuzzy network was evaluated independently on a test set of 267 records before assessing the combined system. The results are summarised in Table V.

TABLE V. CLASSIFICATION ACCURACY OF EACH SYSTEM COMPONENT (TEST SET).

Component	Inputs	Accuracy	Notes
FIS 1 alone	age, BMI	54.7%	No smoker information
FIS 2 alone	smoker, children	91.8%	Smoker is dominant predictor
Merged Network	smoker, age	91.8%	horizontal_merging, 6 rules
Combined System	all four inputs	91.8%	FIS 2 and Merged Network agree on all test records

FIS 1, which models health risk based solely on age and BMI, achieves an accuracy of 54.7%. This relatively low figure is expected and instructive. A cross-tabulation of the dataset demonstrates that no non-smoker subgroup, regardless of age or BMI category, has a mean annual charge exceeding $14,500, which falls below the $16,000 decision boundary. Consequently, FIS 1 lacks the information needed to reliably discriminate between the two charge classes, confirming that physiological indicators alone are insufficient for risk classification in this dataset.

FIS 2, which incorporates smoking status, achieves 91.8% accuracy. The control surface of FIS 2, shown in Fig. 6 (Section III), illustrates the sharp transition in lifestyle_risk as smoking status increases from 0 to 1. It reflects the $23,500 difference in mean charges between the two groups.

B. *Merged Network and Combined System*

The merged network, produced by the horizontal_merging function of RB1 and RB2, maps smoker-age combinations directly to charge categories through the 6 × 2 Boolean rule bases presented in Table IV. Row-index lookup on the test set produces an accuracy of 91.8%, identical to that of FIS 2 alone. This equivalence happens because the merged network encodes the same fundamental rule: smoker status determines the charge category, with age providing the secondary partitioning within RB1, and the smoker boundary is sufficiently discriminating that the age sub-partitioning does not alter any classification on this test set. The agreement between the merged network and FIS 2 across all 267 test records confirms the network design's internal consistency. The combined system, which treats the merged network output as the primary classifier and FIS 2 as a validation signal, therefore also achieves 91.8% accuracy, with both components producing identical predictions on every test record.

C. *Confusion Matrix and Detailed Metrics*

The confusion matrix for the combined system is presented in Table VI, and the obtained performance measures are given in Table VII.

TABLE VI. CONFUSION MATRIX: COMBINED FUZZY NETWORK (TEST SET, N=267).

	Predicted: LowMedium	**Predicted: High**
Actual: LowMedium	195 (TN)	5 (FP)
Actual: High	17 (FN)	50 (TP)

TABLE VII. HIGH-CLASS PERFORMANCE MEASURES: COMBINED FUZZY NETWORK.

Metric	Value	Interpretation
Overall Accuracy	91.8%	245 of 267 test records correctly classified
Precision	90.9%	Of all records predicted High, 90.9% were genuinely High
Recall	74.6%	Of all actual High records, 74.6% were correctly identified
Specificity	97.5%	Of all actual LowMedium records, 97.5% correctly identified
F1 Score	0.820	Harmonic mean of precision and recall

Within 22 misclassified records, 5 are false positives, where the LowMedium charge category was incorrectly predicted as High, and 17 are false negatives, where the High charge category was incorrectly predicted as LowMedium. The false negative group comprises non-smoker patients whose annual charges exceed $16,000 despite belonging to demographic subgroups in which the majority of charges fall below the threshold. As established by the theoretical ceiling analysis in Section III, these 17 records represent the irreducible classification error of any rule-based system on this dataset.

V. EVALUATION

A. *Comparison with Machine Learning Benchmarks*

Table VIII presents the performance of the combined fuzzy network alongside the machine learning benchmarks established in Section II. All methods are evaluated on the same dataset under the same binary classification task, enabling direct and consistent comparison.

TABLE VIII. PERFORMANCE COMPARISON: COMBINED FUZZY NETWORK VERSES MACHINE LEARNING BENCHMARK.

Method	Task	Metric	Performance
Random Forest [9]	Binary classification	Accuracy	83.4%
Gradient Boosting [9]	Binary classification	Accuracy	92.0%
Combined Fuzzy Network (this work)	Binary classification	Accuracy	91.8%
Combined Fuzzy Network (this work)	Binary classification	F1 Score	0.820

The combined fuzzy network achieves 91.8% accuracy, 0.2% lower than the Gradient Boosting benchmark result at 92.0%. The difference between the fuzzy network and Gradient Boosting may correspond to a single misclassified record on the 267-record test set and is not statistically meaningful at this sample size. The fuzzy network outperforms Random Forest by 8.4 percentage points. It is important to note that the 17 false negatives that constitute the remaining error margin are non-smoker records whose charges exceed $16,000 and are not captured by any of the four input variables. They are effectively irreducible under the current feature set. Gradient Boosting's marginal advantage of 0.2% points over the fuzzy network may therefore reflect minor differences in test set composition or threshold selection rather than a substantive performance difference.

B. *Modular Decomposition*

The contrast between FIS 1 (54.7%) and FIS 2 (91.8%) provides a justification for the modular network architecture. A flat rule base combining all four inputs, age, BMI, smoker, and children, would require up to 54 rules ($3 \times 3 \times 2 \times 3$) to achieve complete input coverage. The modular decomposition reduces this to 15 rules across the 2 Mamdani subsystems (9 in FIS 1, 6 in FIS 2), whereas the cascaded network requires only nine Boolean rules in total (6 in RB1, 3 in RB2). This reduction in rule count is consistent with the scalability benefits of fuzzy network architectures.

Furthermore, the low accuracy of FIS 1 in isolation constitutes a meaningful experimental finding. It demonstrates that physiological indicators, age and BMI are secondary predictors of insurance cost on this dataset, whilst smoking status is the primary discriminating variable. This result is consistent with the charge distribution data reported in Section III, where the mean charge differential between smokers and non-smokers ($32,050 versus $8,434) is more than twice the differential attributable to any age or BMI combination within the non-smoker cohort. The modular architecture correctly reflects this asymmetry by assigning smoking status to a dedicated sub-system (FIS 2) that operates independently of the health risk dimension.

C. *Limitations*

The dataset is limited to four retained input variables and does not include potentially predictive features such as chronic disease history, prescription drugs, occupation history, or prior claims record, all of which are routinely employed in commercial insurance pricing models. The binary classification boundary at $16,000 was selected based on the dataset's bimodal charge distribution, and its generalisability may not hold for insurance datasets drawn from different demographic or geographic contexts. Finally, the fuzzy network achieves accuracy comparable to that of Gradient Boosting. Yet, it does not produce a continuous charge estimate and therefore cannot be directly substituted for regression-based pricing models in actuarial applications that require point estimates rather than risk categories.

VI. CONCLUSION

This paper presents a cascaded fuzzy network for classification of medical insurance charge risk, addressing the scalability limitations of conventional flat rule-based systems through a modular decomposition strategy grounded in the fuzzy network architecture of Gegov [6] and his Fuzzy Network Toolbox. Two Mamdani Type-1 subsystems were constructed to model the health and lifestyle-related insurance risks independently, and connected via the horizontal_merging function to produce a 6 x 2 merged rule base that maps directly from smoker–age profiles to binary charge categories.

The combined system achieved 91.8% accuracy on the held-out test set, with a High-class F1 score of 0.820. This performance is only 0.2% lower than the Gradient Boosting benchmark reported by Islam et al. [9] on the same dataset and task, and exceeds the Random Forest benchmark by 8.4%. From the marked disparity in accuracy between FIS 1 (54.7%) and FIS 2 (91.8%), indicates that smoking status is the dominant predictor of insurance cost in this dataset, a finding that informs the network design and is consistent with the observed bimodal charge distribution. The 15-rule Mamdani system achieves accuracy comparable to an ensemble machine learning model whilst maintaining a fully transparent, human-readable rule structure.

Future work may address three limitations identified in this study. First, including additional clinical variables, such as chronic disease history, occupation, or prior claims record could improve recall for the High-risk class, which remains the primary source of classification error. Second, the binary classification boundary at $16,000 is specific to this dataset's charge distribution and would require recalibration for application to insurance data from different demographic or geographic contexts. Third, extending the network architecture to incorporate output merging or vertical merging operations from the Fuzzy Network Toolbox would enable multi-class charge categorisation, producing a more granular risk stratification output suitable for actuarial pricing applications.

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